Recurrence relations

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First-order linear recurrences

For $a_n = pa_{n-1} + q$, p≠1 gives fixed point $a^* = q/(1-p)$. Subtracting it yields $b_n = pb_{n-1}$, hence

$$a_n = p^{n-n_0} a_{n_0} + q\frac{1-p^{n-n_0}}{1-p}.$$

For p=1, $a_n = a_{n_0} + (n-n_0)q$. Specify the initial index. Integer powers use $p^0=1$, including the empty-product convention $0^0=1$, without asserting a two-variable limit.

Unrolling for $n>n_0$ gives $a_n = p^2 a_{n-2} + pq + q = \cdots = p^{n-n_0} a_{n_0} + q(1 + p + \cdots + p^{n-n_0-1})$. The geometric sum gives the first formula for p≠1. For p=1 there are $n-n_0$ additions of q; p=0 gives $a_n=q$ after the initial index. Check the initial term, next term, and original recurrence. The fixed-point substitution is a convenient homogenization of the same unrolling proof.

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General forcing terms

For $a_n = pa_{n-1} + f(n)$, p≠0 allows the integrating factor

$$p^{-n}a_n - p^{-(n-1)}a_{n-1} = p^{-n}f(n).$$

Telescoping gives

$$a_n = p^{n-n_0}a_{n_0} + \sum_{k=n_0+1}^{n} p^{n-k}f(k).$$

The final formula no longer divides by p, so remains defined for p=0,1 and can be verified independently by substitution. Only the integrating-factor derivation needed p≠0. Negative p requires retaining parity signs in integer powers, rather than assuming positive-term geometric intuition.

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Variable coefficients and applications

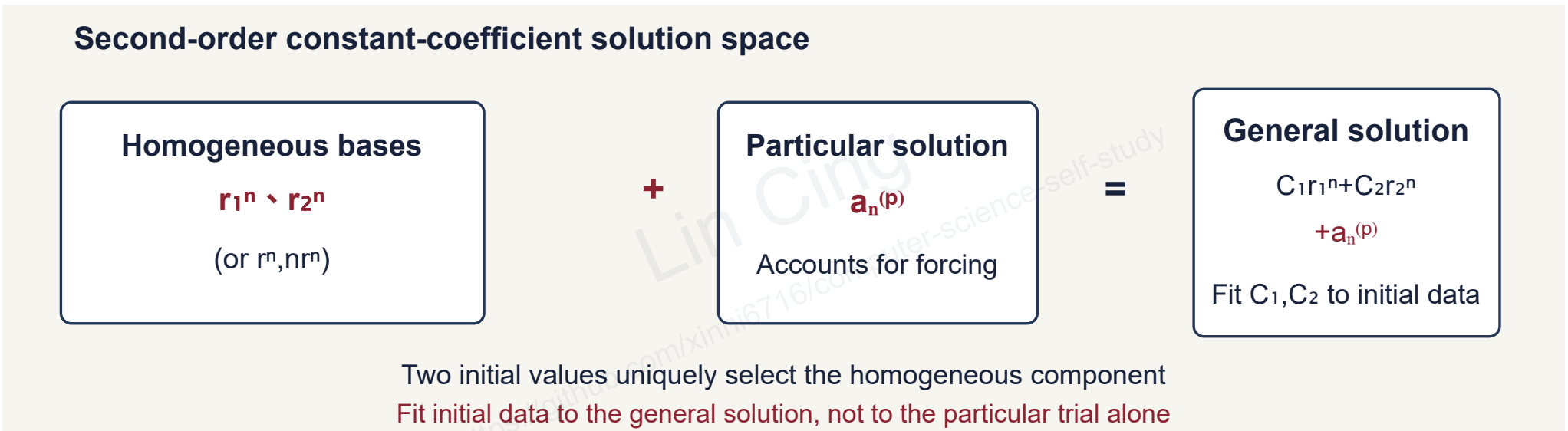
For $a_0=1$ and $a_n=na_{n-1}$, unrolling gives $a_n=n(n-1)\dots 1a_0=n!$. With $a_0=0$ every term is instead zero, showing why initial data are part of a recurrence specification.

Worst-case bubble-sort comparisons satisfy $h_1=0, h_n=h_{n-1}+n-1$: the first pass uses $n-1$ comparisons and leaves $n-1$ elements. Thus $h_n = \sum_{j=1}^{n-1} j = n(n-1)/2 = \Theta(n^2)$. This counts comparisons, not identical elapsed time on every machine; memory hierarchy and branch prediction still matter. Variable coefficients do not directly admit constant-coefficient characteristic roots; try products, telescoping, or substitutions and verify the actual per-round states.

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Distinct characteristic roots

For $a_n + Aa_{n-1} + Ba_{n-2} = 0$, n≥2, trial r^n gives characteristic polynomial r^2+Ar+B. Distinct roots give $a_n = C_1 r_1^n + C_2 r_2^n$, with two initial conditions fixing constants. Check zero roots and the index domain separately.



Substitution gives $r^{n-2}(r^2 + Ar + B) = 0$, so a nonzero root supplies a solution. For nonzero distinct roots,

Distinct characteristic roots (continued)

$$\det \begin{pmatrix} r_1^n & r_2^n \\ r_1^{n+1} & r_2^{n+1} \end{pmatrix} = r_1^n r_2^n (r_2 - r_1) \neq 0.$$

They are independent. Every solution is uniquely determined by a_0,a_1, so the solution space has dimension two. The initial equations $C_1+C_2=a_0$,$C_1r_1+C_2r_2=a_1$ have determinant $r_2-r_1\neq 0$ and uniquely fit all solutions. Normalize the recurrence's signs before forming its polynomial.

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Repeated, higher-order, and complex roots

A repeated nonzero root ρ gives $a_n = (C_1 + C_2 n)\rho^n$. To prove the second basis, use A=−2ρ,B=ρ²:

$$n\rho^n + A(n-1)\rho^{n-1} + B(n-2)\rho^{n-2} = \rho^{n-2}[n\rho^2 - 2\rho^2(n-1) + \rho^2(n-2)] = 0.$$

The initial-value matrix $\begin{pmatrix} 1 & 0 \\ \rho & \rho \end{pmatrix}$ has determinant ρ≠0, proving independence and completeness. If ρ=0, the recurrence only forces a_n=0 for n≥2, leaving a_0,a_1 arbitrary. The pair 0^n,$n0^n$ is not a basis.

For a nonzero root r of multiplicity m in an order-k recurrence, use r^n,nr^n,…,$n^{m-1}r^n$. Handle each root's multiplicity separately; the total basis count must equal k. For zero multiplicity m_0 use prefix sequences δ_0,…,δ_{m0-1}, where $\delta_j(n)$=1 iff n=j. Conjugate complex roots can be paired into real trigonometric solutions. Fibonacci's roots are irrational real numbers, not nonreal conjugates.

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General proof for higher-order solutions

Write the forward recurrence as P(E)a=0, with $Ea_n=a_{n+1}$, monic degree-k P, and $n \geq 0$. The first k arbitrary values determine the rest uniquely, so dimension is k. For nonzero r and polynomial q,

$$(E - r)(r^n q(n)) = r^{n+1} \Delta q(n).$$

The difference operator lowers positive degree by one and kills constants. Thus multiplicity m makes P(E) annihilate $r^n n^j$ for $j<m$. For another root $\lambda \neq r$,

$$(E - \lambda)(r^n q(n)) = r^n [(r - \lambda) q(n) + r \Delta q(n)].$$

On polynomials of degree below m this map is triangular with nonzero diagonal $r-\lambda$, hence invertible.

In a proposed relation among bases, apply all other roots' factors to isolate the chosen r. Other terms vanish; the surviving polynomial has undergone only invertible maps and must vanish at every nonnegative integer, hence is the zero polynomial. Repeat for each root to prove independence.

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General proof for higher-order solutions (continued)

The kernel of E^{m_0} is precisely sequences supported at indices 0,…,m₀−1. Each E−λ for nonzero λ is triangular on this space with nonzero diagonal −λ. The same isolation argument handles the zero-root prefix bases. All k bases are independent and hence span the full solution space. Factor P over the complex numbers; real coefficients permit conjugate pairing. Roots $\rho e^{\pm i\theta}$ give real solutions $\rho^n(A\cos n\theta + B\sin n\theta)$, with real A,B fitted from real data. Repeated roots require additional independent polynomial factors.

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Nonhomogeneous recurrences and resonance

Linearity gives $a_n = a_n^{(h)} + a_n^{(p)}$. First solve the homogeneous equation, choose a particular trial matching the forcing type, substitute and compare coefficients, then add the homogeneous solution and fit initial data to the total. If its base r is a root of multiplicity m, multiply the trial by n^m.

Forcing	Trial before resonance
Constant	Constant
Polynomial degree d	General polynomial degree d
Cr^n	Ar^n
Degree-d polynomial times r^n	General degree-d polynomial times r^n
$A\cos(n\theta)+B\sin(n\theta)$	Both sine and cosine at that frequency

A corresponding root collision multiplies the whole trial by n^m, including conjugate collisions for sinusoidal forcing.

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Nonhomogeneous recurrences and resonance (continued)

For $a_n=3a_{n-1}+4n, a_0=2$, the homogeneous solution is $C3^n$. Trial $An+B$ gives $An+B=(3A+4)n+(3B-3A)$, so $A=-2, B=-3$. Then $a_n=C3^n-2n-3$, and $a_0=2$ gives $C=5$. Thus $a_n=5\cdot 3^n-2n-3$, with terms $2,10,38$; $3\cdot 10+8=38$.

In $a_n-a_{n-1}=2^n$, base two does not collide with root one. In $a_n-2a_{n-1}=2^n$ it does, requiring $Cn2^n$. For $a_n=2a_{n-1}+5\cdot 2^n, a_0=3$, substitute $An2^n$:

$$An2^n = A(n-1)2^n + 5\cdot 2^n, \qquad A = 5.$$

Thus $a_n=(C+5n)2^n$ and $C=3$. Terms $a_1=16=2\cdot 3+10$ and $a_2=52=2\cdot 16+20$ check the result.

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Nonhomogeneous recurrences and resonance (continued)

Check the particular trial for resonance

Forcing term

$f(n)=A\,r^n$

First try $B\,r^n$

→

Is r a homogeneous root?

No: trial $B\,r^n$

Yes: multiplicity m

Trial $B\,n^m\,r^n$

Example $a_n=2a_{n-1}+5\cdot 2^n$

Root 2, multiplicity m=1

Trial $An2^n$

At resonance, $B\,r^n$ cancels on substitution and cannot produce the forcing

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Nonhomogeneous recurrences and resonance (continued)

Superposition requires linearity; it does not justify adding particular solutions for nonlinear equations. The resonance rule follows from difference operators. For nonzero r of multiplicity s, $(E-r)^s$ lowers polynomial degree s times; other factors preserve degree and are invertible on each fixed-degree space. Taking $Q(n)=n^sH(n)$, deg $H \leq d$, the map Δ^s sends n^{s+j} to degree j with nonzero leading coefficient. This is an invertible triangular map to degree $\leq d$ polynomials, proving existence for every degree-d polynomial times r^n forcing. At s=0 no n-factor is needed. Nonzero r, constant coefficients, and characteristic zero are required.

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Generating functions and rationality

Finite-order homogeneous constant-coefficient recurrences give Q(x)A(x)=P(x), Q(0)≠0, by multiplying by x^n and summing. Thus the OGF is rational. Rational forcing OGFs preserve rationality. Linearity alone, especially with variable coefficients, does not suffice.

Conversely, if $A = P/Q, Q = 1 + q_1 x + \cdots + q_d x^d$, coefficient comparison past deg P gives $a_n + q_1 a_{n-1} + \cdots + q_d a_{n-d} = 0$. An eventually constant-coefficient recurrence with finitely many exceptions conversely yields a rational OGF. These are formal statements independent of convergence.

For $a_n + C_1 a_{n-1} + C_2 a_{n-2} = f_n$, n≥2, put $F(x) = \sum_{n \geq 2} f_n x^n$. The sums are $A - a_0 - a_1 x$, $x(A - a_0)$,$x^2 A$. Therefore

$$(1 + C_1 x + C_2 x^2) A = a_0 + (a_1 + C_1 a_0) x + F(x).$$

The denominator reflects recurrence coefficients; initial values and forcing determine the numerator.

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Generating functions and rationality (continued)

For $a_0=1, a_1=4, a_n=5a_{n-1}-6a_{n-2}$, roots 2,3 give $C_1+C_2=1, 2C_1+3C_2=4$, hence $C_1=-1, C_2=2$ and $a_n=-2^n+2\cdot 3^n$. The OGF route gives

$$(A-1-4x)-5x(A-1)+6x^2A=0, \qquad A=\frac{1-x}{(1-2x)(1-3x)}=-\frac{1}{1-2x}+\frac{2}{1-3x}.$$

Both methods retain the same numerator and constants.

For $a_0=1, a_n=3a_{n-1}+2$, $A-1=3xA+2x/(1-x)$, giving $A=(1+x)/((1-x)(1-3x))$. Partial fractions give $a_n=2\cdot 3^n-1$, with 1,5,17,53.

For $h_0=0, h_n=h_{n-1}+1/n$, linearity still holds but

$$(1-x)H=\sum_{n\geq 1} x^n/n=-\log(1-x), \qquad H=-\log(1-x)/(1-x).$$

The initial terms are 0,1,3/2,11/6,25/12. Differentiating the forcing series gives $1/(1-x)$; its zero constant fixes $-\log(1-x)$.

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Generating functions and rationality (continued)

To prove nonrationality, suppose H rational. Then R=(1−x)H=−log(1−x) is rational. At x=1 a rational function either has a finite limit or a pole of finite integer order m≥1. Set t=−log(1−x). As x→1⁻, t→∞ but $t/(1-x)^{-m} = te^{-mt} \to 0$ for every fixed m. Logarithmic divergence is neither finite nor any pole order, contradiction. A formal rational representation would have the same Taylor expansion and analytic function, so is excluded too. Both examples require initial corrections; rationality depends on coefficients and forcing, not just linearity.

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Nonlinear recurrences

Equations such as $a_{n+1}=a_n+a_n^2$, or recurrences with products and quotients, usually have no fixed linear characteristic polynomial. Seek monotonicity, bounds, fixed points, or a reversible substitution. The change $b_n=\log a_n$ requires $a_n>0$ and converts products into sums. A nonlinear problem may ask for asymptotic bounds or invariants rather than a closed form; specify the target.

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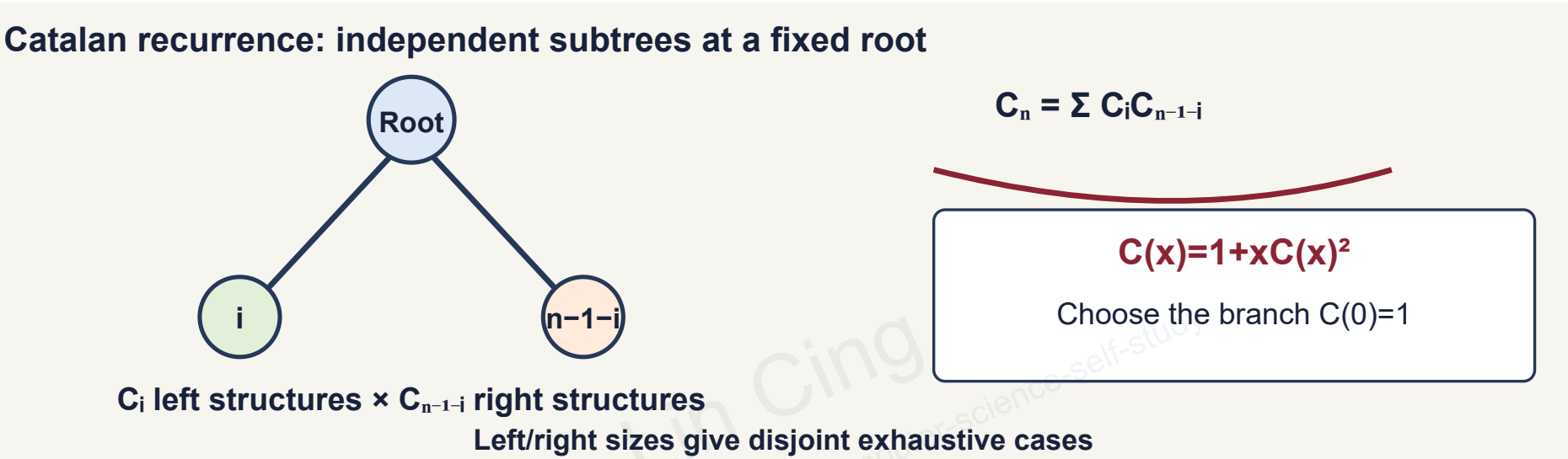
Catalan convolution and a nonlinear closed form

Let C_n count ordered rooted binary trees on n vertices, with $C_0=1$ for the empty tree. A root uniquely joins left and right subtrees of sizes i,n−1−i, giving

$$C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i}.$$

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Catalan convolution and a nonlinear closed form (continued)



Summing gives $C-1=xC^2$, so $xC^2-C+1=0$. The branch with constant one is $C = (1 - \sqrt{1-4x})/(2x)$.

Construct the formal square root directly. Let $Y(z) = \sum_{j\geq 0} \binom{1/2}{j} z^j$, with descending-product binomials. Coefficient recurrence $(j+1)c_{j+1}=(1/2-j)c_j$ gives $(1+z)Y'=Y/2, Y(0)=1$. For $F=Y^2$, $(1+z)F'=F, F(0)=1$; coefficient comparison uniquely yields $F=1+z$. Thus Y is the square root with constant one, not the negative branch. Substitute $z=-4x$ and simplify descending products:

Catalan convolution and a nonlinear closed form (continued)

$$\sqrt{1-4x} = 1 - 2\sum_{k\geq 1} \frac{1}{k}\binom{2k-2}{k-1} x^k.$$

Substitution and n=k−1 give $C_n = \binom{2n}{n}/(n+1)$, with 1,1,2,5,14,42. At n=3 the five ordered trees biject with five stack outputs of {1,2,3}. Identifying left/right subtrees changes the object and invalidates the same count.

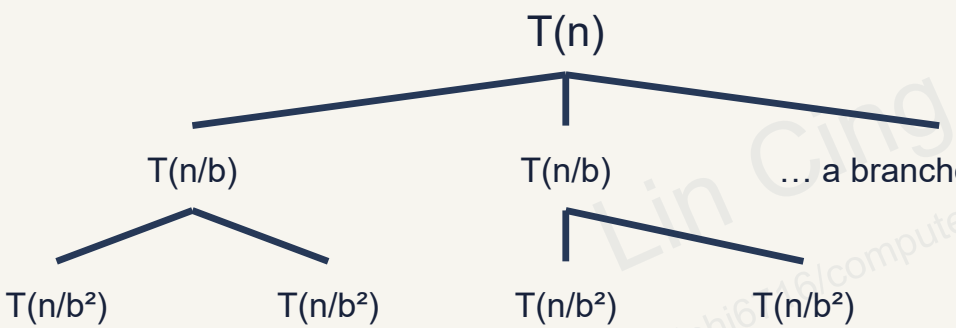
The nonlinear sequence $a_0=1, a_{n+1}=a_n+a_n^2$ gives 1,2,6,42,… and is not the Catalan recurrence. Seek transformations, convolution, or growth invariants rather than characteristic roots.

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Divide-and-conquer recurrences

For $T(n) = aT(\lceil n/b \rceil) + f(n)$, handle exact floors/ceilings and ensure subproblems strictly decrease past the base threshold. A polynomial-cost master theorem compares a with b^d for $aT(n/b)+n^d$, identifying leaves, equal-cost layers, or the root as dominant.

Divide-and-conquer cost levels



Per level

$a^j f(n/b^j)$

Compare growth rates for cost

Branching versus decay per level

Shrinking subproblems and complete base cases give finite depth

Fix integers $a \geq 1, b \geq 2, c > 0$ and $n = b^k$. For $f(n) = af(n/b) + c, f(1) = c$,

$$f(b^k) = a^k f(1) + c(1 + a + \cdots + a^{k-1}).$$

Divide-and-conquer recurrences (continued)

If a=1, this is c(k+1)=Θ(log_b n). If a≥2 the sum is (aᵏ−1)/(a−1), yielding Θ(aᵏ)=Θ($n^{\log_b a}$). Generally,

$$f(b^k) = a^k f(1) + \sum_{j=0}^{k-1} a^j h(b^{k-j}).$$

For fixed d≥0,h(n)=Θ(n^d)>0,f(1)=Θ(1)>0, the layer ratio a/b^d gives $\Theta(n^{\log_b a})$ when d<log_b a, Θ(n^dlog n) at equality, and Θ(n^d) when d>log_b a. This proof is on powers of b. General integers require floor/ceiling and monotonicity bounds between adjacent powers.

For T(n)=2T(n/2)+n/$\log_2$n on n=2^k≥2, a polynomial-forcing theorem cannot replace n/log n by n. Level j costs n/(k−j), so

$$T(2^k) = 2^k T(1) + 2^k \sum_{j=0}^{k-1} \frac{1}{k-j} = n(T(1) + H_k) = \Theta(n \log\log n).$$

Use a separate positive T(1); do not divide by $\log_2 1$. At n=16,T(1)=1, internal levels cost 4,16/3,8,16 and leaves cost 16, totaling 148/3=16(1+H_4). A recurrence that still calls itself at n=1 has no termination proof.

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Master-theorem examples and limits

For positive constant bases, $2T(n/2)+n$ gives $\Theta(n \log n)$, $4T(n/2)+n$ gives $\Theta(n^2)$, and $2T(n/4)+n$ gives $\Theta(n)$. Recursion-tree layer sums verify each. Nonpolynomial forcing or unsuitable a,b and division conditions require a broader theorem, recursion trees, Akra–Bazzi, or direct induction. A Θ claim requires both upper and lower bounds; $O(n \log n)$ alone does not complete a tight analysis.

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Choosing a recurrence method

Use iteration or integrating factors for first order; characteristic roots for homogeneous constant-coefficient linear equations; homogeneous plus particular solutions with resonance checks for forcing; generating functions for coefficient extraction; transformations, invariants, or bounds for nonlinear equations; and exact size recurrences followed by justified tree/theorem/induction analysis for divide and conquer. Characteristic roots outside their linear constant-coefficient family are a mathematical error, not a stylistic choice.

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Knockout tournaments

With $n = 2^m$ entrants, each game eliminates one player. Every player except the champion is eliminated once, so $n-1$ games suffice and are necessary. Equivalently $T(1)=0, T(n)=2T(n/2)+1$ unrolls to $n-1$. One argument counts eliminations, the other solves a recurrence; agreement checks the base and merge-cost meanings.

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Coupled recurrences and simultaneous generating functions

For fixed matrix M and $y_n=My_{n-1}+f_n$, let $G=\sum_{n\ge 0} y_n x^n$, $F=\sum_{n\ge 1} f_n x^n$. Then

$$G-y_0=xMG+F, \qquad G=(I-xM)^{-1}(y_0+F).$$

No invertibility of M or analytic convergence is needed. The formal series $\Sigma x^k M^k$ multiplies $I-xM$ to I by cancellation of every positive-degree coefficient. If F has rational components, the adjugate formula and determinant constant one make G's components rational.

For $a_n=2a_{n-1}+b_{n-1}, b_n=a_{n-1}+b_{n-1}, a_0=1, b_0=0$, use the same old pair in both updates.

n	Old pair	New a_n	New b_n
1	(1,0)	2 • 1+0=2	1+0=1
2	(2,1)	2 • 2+1=5	2+1=3
3	(5,3)	2 • 5+3=13	5+3=8
4	(13,8)	2 • 13+8=34	13+8=21

The OGF system is

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Coupled recurrences and simultaneous generating functions (continued)

$$\begin{pmatrix} 1-2x & -x \\ -x & 1-x \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad A = \frac{1-x}{1-3x+x^2}, \quad B = \frac{x}{1-3x+x^2}.$$

Both components satisfy $u_n = 3u_{n-1} - u_{n-2}$ but have distinct initial values. With $r_\pm = (3 \pm \sqrt{5})/2$, partial fractions give

$$b_n = \frac{r_+^n - r_-^n}{\sqrt{5}}, \qquad a_n = \frac{(r_+ - 1)r_+^n + (1 - r_-)r_-^n}{\sqrt{5}}.$$

At n=0,1 these give (1,0),(2,1); uniqueness of each second-order recurrence then proves agreement for all terms. Elimination must preserve each component's own initial values.

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Boundaries and verification

Check initial values and substitute into the original recurrence. Verify index floors/ceilings, eventual scope of asymptotic bounds, stopping conditions, and the operation model. Exercises and solutions provide further derivations. Closed forms are one objective; lower-order terms can matter to actual cost or memory and cannot always be discarded.

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References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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